

OpenCPU SDK

Paho MQTT Embedded API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

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Audience

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Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Niu Sulong

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document provides API descriptions for implementing MQTT connections based on the open-source library `paho.mqtt.embedded-c-master`.

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2 APIs

The following interface function definitions can be found in MQTTClient.h and MQTTFreeRTOS.h.

2.1 MQTTIsConnected

Format	int MQTTIsConnected(MQTTClient* client)
Description	Checks the MQTT connection status.
Parameters	client: MQTT client instance.
Return Value	Successful: 0 Failure: other error code
Remarks	None

2.2 NetworkInit

Format	void NetworkInit(Network* n)
Description	Initializes the network.
Parameters	n: Network connection parameters.
Return Value	None
Remarks	None

2.3 MQTTClientInit

Format	void MQTTClientInit(MQTTClient* c, Network* network, unsigned int command_timeout_ms, unsigned char* sendbuf, size_t sendbuf_size, unsigned char* readbuf, size_t readbuf_size)
Description	Initializes an MQTT client instance.
Parameters	c: MQTT client instance. network: Network connection parameter. command_timeout_ms: Command timeout. sendbuf: Buffer for sending data.

	sendbuf_size: Size of the sending buffer. readbuf: Buffer for receiving data. readbuf_size: Size of the receiving buffer.
Return Value	None
Remarks	None

2.4 NetworkConnect

Format	int NetworkConnect(Network* n, char* addr, int port)
Description	Connects to a network.
Parameters	client: MQTT client instance.
Return Value	n: Network connection parameters. addr: server URL port: server port number.
Remarks	None

2.5 MQTTConnect

Format	Int MQTTConnect(MQTTClient* c, MQTTPacket_connectData* options)
Description	Connects to an MQTT broker.
Parameters	c: MQTT client instance. options: Connection options, including client identifier, username, password, keep-alive time, etc.
Return Value	Success: 0
Remarks	Assign a value to the defaultMessageHandler member in the c structure before connecting.

2.6 MQTTPublish

Format	int nwy_MQTTPublish(MQTTClient* c, const char* topicName, MQTTMessage* message)
Description	Publishes a message to an MQTT topic.
Parameters	topicName: Name of the topic. message: Message body, including content, length, and retain flags.

Return Value	Successful: 0 Failure: other error codes
Remarks	None

2.7 MQTTSubscribe

Format	int MQTTSubscribe(MQTTClient* c, const char* topicFilter, enum QoS qos, messageHandler messageHandler)
Description	Subscribes to an MQTT topic and registers a message handler.
Parameters	topicFilter: Name of the topic. qos: Quality of Service level. messageHandler: Callback function to handle incoming messages.
Return Value	Successful: 0 Failure: other error code
Remarks	None

2.8 MQTTDisconnect

Format	int MQTTDisconnect(MQTTClient* c)
Description	Disconnects from the MQTT broker and releases resources.
Parameters	c: MQTT client instance.
Return Value	Successful: 0 Failure: other error code
Remarks	None

2.9 NetworkDisconnect

Format	void NetworkDisconnect(Network* n)
Description	Disconnects from the network.
Parameters	n: Network parameters.
Return Value	None
Remarks	None

2.10 MQTTYield

Format	int MQTTYield(MQTTClient* c, int timeout_ms)
Description	Receives network packets and dispatches messages to user-defined callback functions.
Parameters	c: MQTT client instance. timeout_ms: Packet reception timeout in milliseconds.
Return Value	Successful: 0 Failure: other error code
Remarks	None